

JavaScript Introduction – Simple Hinglish Summary

Source: <https://javascript.info/intro>

1. JavaScript kya hai?

- JavaScript ko banaya gaya tha **web pages ko "alive" banane ke liye** (click, animation, dynamic content).
- Iske programs ko **scripts** kehte hain. Ye plain text me hote hain, alag se compile karne ki zarurat nahi.
- **Java aur JavaScript alag hain.** Pehle iska naam "LiveScript" tha, Java popular tha isliye naam JavaScript rakh diya. Iski official spec ka naam **ECMAScript** hai.

2. JavaScript Engine

- JS chalane ke liye ek **engine** chahiye. Browser me pehle se hota hai.
 - **V8** → Chrome, Edge, Opera
 - **SpiderMonkey** → Firefox
 - **JavaScriptCore** → Safari
- Engine ka kaam: script **parse** karo → **machine code me compile** karo → **run** karo (fast).

3. Browser me JS kya kar sakta hai?

- HTML/CSS badalna, naya content add karna.
- Click, mouse, keyboard jaise user actions par react karna.
- Server ko requests bhejna (AJAX), files upload/download.
- Cookies aur local storage me data save karna.

4. Browser me JS kya NAHI kar sakta? (Safety ke liye)

- Hard disk ki files ko seedha read/write nahi kar sakta (user permission chahiye).
- Camera/mic bina permission ke use nahi ho sakta.
- Ek tab dusre tab ka data nahi dekh sakta agar site alag ho (**Same Origin Policy**).
- Dusri site se data lena mushkil hai, jab tak wo site allow na kare.
- Ye limits sirf browser me hain. **Node.js (server) me ye limits nahi hoti.**

5. JavaScript special kyu hai?

Teen badi wajah:

1. HTML/CSS ke saath **full integration**
2. **Simple cheezein simple tarike se** ho jati hain
3. **Sab major browsers me by default supported**

Aaj JS se server, mobile apps, etc. bhi ban sakte hain.

6. JavaScript ke "upar" ki languages

Kuch languages **transpile** hoke JavaScript ban jati hain:

- **TypeScript** - strict data types (Microsoft)
- **CoffeeScript** - chhota syntax
- **Flow** - data types (Facebook)
- **Dart** - Google ki language
- **Brython** - Python se JS
- **Kotlin** - modern aur safe language

Inhe use karne ke liye bhi **JavaScript aani chahiye**.

Quick Summary

- JS pehle sirf browser ke liye thi, ab har jagah use hoti hai.
- Ye sabse zyada use hone wali browser language hai, HTML/CSS ke saath perfect fit.
- Pehle JS seekho, phir TypeScript jaisi languages dekho.